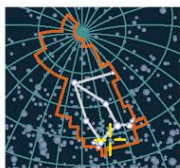


PULSATING VARIABLE

Mu Cephei



DISTANCE FROM SUN
5,258 light-years

MAGNITUDE 4

SPECTRAL TYPE M2

CEPHEUS

Mu (μ) Cephei, or the Garnet Star, is also known as Herschel's Garnet Star, after the pioneering German-born astronomer William Herschel, who first described its distinctive red color and noted its resemblance to the precious stone garnet. Mu Cephei is one of the most luminous stars in the Milky Way, outshining the Sun by a factor of more than 200,000. A red supergiant, it is also one of the largest stars that can be seen with the naked eye. Its great size means that if placed in the Sun's position at the center of the solar system, its outer layers would fall between Jupiter and Saturn. As with most large supergiants, Mu Cephei is an unstable star, expanding and contracting in diameter with a corresponding variation in magnitude. It is classed as a semiregular supergiant variable with a spectral type of M2 and a magnitude varying between 3.43 and 5.1. It has two periods of variation (730 and 4,400 days) overlaid on one another. The pulsations of Mu Cephei, caused by internal absorption and release of energy, have thrown off the outer

layers of the star's atmosphere, creating concentric shells of dust and gas around it. Observations have also shown that Mu Cephei is surrounded by a sphere of water vapor. Mu Cephei probably started its life as a star of around 20 solar masses. Typically for such a high-mass star, it has evolved very rapidly, and we are seeing it as it hurtles headlong toward

the end of its short life. One day soon (on an astronomical timescale), Mu Cephei will erupt in a cataclysmic supernova, after which only the core will remain, ending its days as a neutron star or black hole.

VARIABLE JEWEL

Known for its distinctive color, Mu Cephei, or the Garnet Star, is the bright reddish-orange star at the top left of the image. It is pictured above the red emission nebula IC 1396 (see p.243).

NOVA

RS Ophiuchi



DISTANCE FROM SUN
2,000–5,000 light-years

MAGNITUDE 12.5

SPECTRAL TYPE M2

OPHIUCHUS

RS Ophiuchi is a recurrent nova (see p.282), having been witnessed erupting in 1898, 1933, 1958, 1967, 1985, and—most recently—in 2006. During its periods of normality, it shines at magnitude 12.5, but during outbursts, it has reached 4th magnitude. While RS Ophiuchi is usually invisible to the naked eye, during outbursts, it can be seen in the night sky without a telescope. RS Ophiuchi is classed as a cataclysmic variable—a binary system consisting of a giant star shedding material and a dwarf companion receiving the material. Eventually, a thermonuclear explosion is triggered on the surface of the dwarf, resulting in the ejection of a shell and an increase in brightness. RS Ophiuchi is constantly monitored by amateur astronomers, and the American Association of Variable Star Observers has more than 30,000 observations in its database.

IRREGULAR VARIABLE

R Coronae Borealis



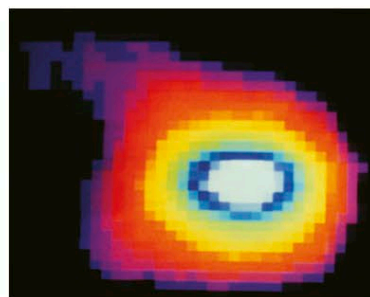
DISTANCE FROM SUN
6,037 light-years

MAGNITUDE 5.9

SPECTRAL TYPE G0

CORONA BOREALIS

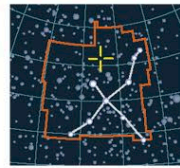
The prototype of a class of variable stars, R Corona Borealis (R CrB) drops in magnitude from 5.9 to 14.4 at irregular intervals. There are two theories for this variation. One is that an orbiting dust cloud obscures R CrB when it passes in front of the star. The other is that R CrB ejects material, which obscures the light the star emits, before being blown away.



FALSE-COLOR INFRARED IMAGE

NOVA

Nova Cygni 1992



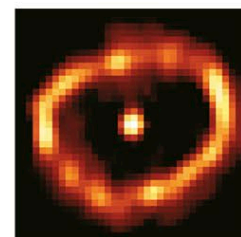
DISTANCE FROM SUN
10,430 light-years

MAGNITUDE 4.3

SPECTRAL TYPE Q

CYGNUS

Nova Cygni 1992, a cataclysmic binary, was discovered on the night of February 18–19, 1992, shining with a magnitude of 7.2 at a location where there should have been no such star. The discoverer, Peter Collins (see p.80), alerted astronomical authorities, and soon a whole range of instruments—both ground-based and space-borne (including the Voyager spacecraft)—were observing it at a variety of wavelengths. Over the next few days, the nova continued to brighten to magnitude 4.3, making Nova Cygni 1992 not only the first nova to be observed so extensively, but also the first to be thoroughly observed before it had reached its peak. The nova eruption was the result of material falling from one star onto a white dwarf, triggering an explosion and the



BRIGHT NOVA

Nova Cygni 1992 was one of the brightest nova to be witnessed erupting in recent history. Targeted by some of the most powerful telescopes in the world, it could be seen at its brightest with the naked eye.

HOT BUBBLE

This Hubble Space Telescope photograph reveals the irregularly shaped bubble of hot stellar material blasted into space by the eruption of Nova Cygni 1992.

ejection of a shell of material. The Hubble Space Telescope observed the system in 1993, detecting the ring thrown out by the binary system and also an unusual barlike structure across the middle of the ring, the origin of which is unknown.